GMI German-Malaysian Institute

DIPLOMA PROGRAMME

PRACTICAL TEST 2

Academic Period : JULY 2026

Subject	**:**	**CRYPTOGRAPHY**
Code	**:**	**CBS 2373**
Duration	**:**	**3 HOURS**
Total Marks	**:**	**50 MARKS**

GENERAL INSTRUCTIONS:

1. Write down clearly (with capital letters) your name, trade, group and the date of examination at the bottom of this page.
2. Do not open this question booklet until you are told to do so by the invigilator.
3. Read the specific instructions and questions carefully before answering.

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Student's I/D No.	**:**	CBS24070208
Semester/Trade/Group	**:**	SEM 5 / DCBS / CBS 1
Date	**:**	28/10/2026

No	Assessment Criteria	CLO	PO	Level of Diff	Full Marks	Score	Marks	Comment
1.	RSA message (text) - To generate RSA keys, and to encrypt and decrypt a text message by using Excel (include the calculation)	2	4	P3	15			
2.	Digital signature - To achieve authentication and integrity by using digital signature when the private key is given, and to detect a changed message (include the calculation)	2	4	P3	20			
3.	Digital certificate - To verify digital certificates from a Certificate Authority (CA) and to detect a forged certificate (include the calculation)	2	4	P3	15			
				Total	**50**			
				Total Marks		/50		
Total			**PO4**		/50			
				Signature		Jumamelissa Sani		

Learning Objectives:

1. To generate RSA public key and private key
2. To encrypt and decrypt:
    a. RSA message (text), letter by letter
    b. Digital signature with message digest (SD17) and RSA
    c. Digital certificate signed by a Certificate Authority (CA)

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To do this Practical Test, you should open the Microsoft Excel and start to follow the question given. These tasks are based on RSA public key & private key, message digest, digital signature and digital certificate.

You are required to submit the answer in word document to PDF FORMAT and EXCEL FILE with mod calculation.

All answers are based on the Excel solution given.

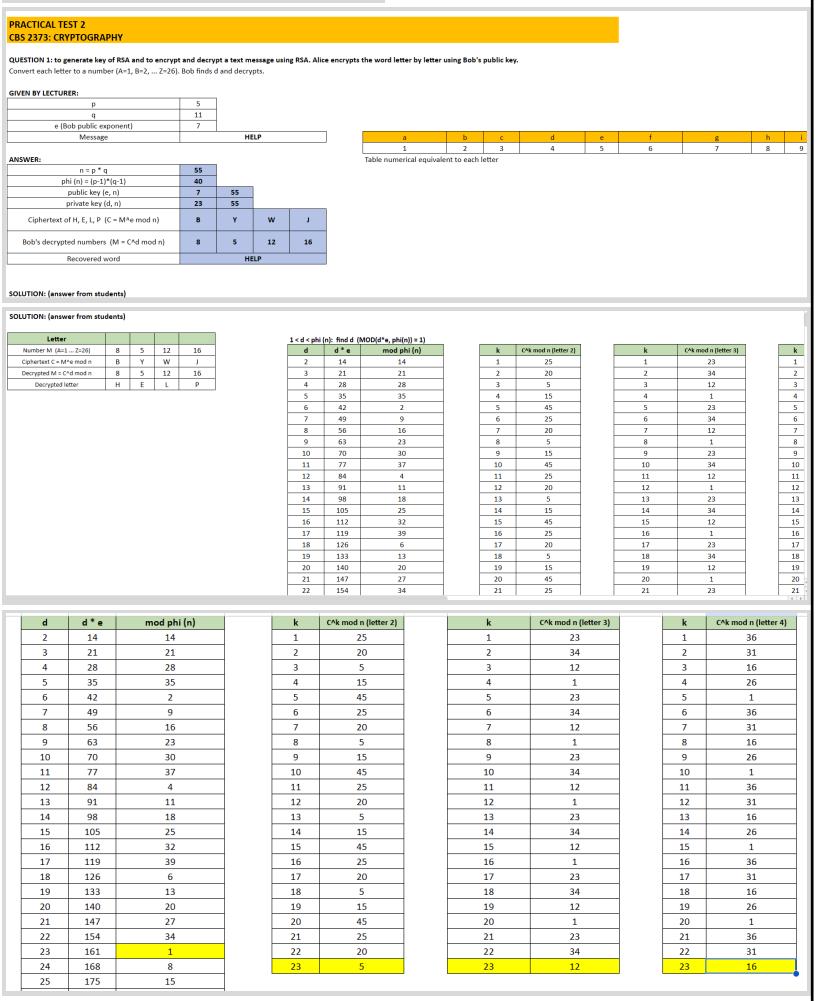
Tasks	Questions	Marks

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RSA message (text): To generate RSA keys, and to encrypt and decrypt a text message by using Excel (include the calculation)	**QUESTION 1:** Alice encrypts the word below letter by letter using Bob's public key (A=1, B=2, ... Z=26). Find Bob's keys and give the encryption and decryption answer by using **Excel including the calculation**. **(keys 5 marks + encryption 5 marks + decryption 5 marks)**	**15**

p	5
q	11
e	7
Message	HELP
n and phi (n)	?
Private key d	?
Encryption (H, E, L, P)	?
Decryption (numbers and word)	?

Screen shot the Excel result:



PRACTICAL TEST 2
CBS 2373: CRYPTOGRAPHY

QUESTION 1: to generate key of RSA and to encrypt and decrypt a text message using RSA. Alice encrypts the word letter by letter using Bob's public key.
Convert each letter to a number (A=1, B=2, ... Z=26). Bob finds d and decrypts.

GIVEN BY LECTURER:

p	5
q	11
e (Bob public exponent)	7
Message	HELP

a	b	c	d	e	f	g	h	i
1	2	3	4	5	6	7	8	9

Table numerical equivalent to each letter

ANSWER:

n = p * q	55			
phi (n) = (p-1)*(q-1)	40			
public key (e, n)	7	55		
private key (d, n)	23	55		
Ciphertext of H, E, L, P (C = M^e mod n)	8	Y	W	J
Bob's decrypted numbers (M = C^d mod n)	8	5	12	16
Recovered word	HELP			

SOLUTION: (answer from students)

SOLUTION: (answer from students)

Letter				
Number M (A=1 ... Z=26)	8	5	12	16
Ciphertext C = M^e mod n	8	Y	W	J
Decrypted M = C^d mod n	8	5	12	16
Decrypted letter	H	E	L	P

1 < d < phi (n): find d (MOD(d*e, phi(n)) = 1)

d	d * e	mod phi (n)
2	14	14
3	21	21
4	28	28
5	35	35
6	42	2
7	49	9
8	56	16
9	63	23
10	70	30
11	77	37
12	84	4
13	91	11
14	98	18
15	105	25
16	112	32
17	119	39
18	126	6
19	133	13
20	140	20
21	147	27
22	154	34

k	C^k mod n (letter 2)
1	25
2	20
3	5
4	15
5	45
6	25
7	20
8	5
9	15
10	45
11	25
12	20
13	5
14	15
15	45
16	25
17	20
18	5
19	15
20	45
21	25

k	C^k mod n (letter 3)
1	23
2	34
3	12
4	1
5	23
6	34
7	12
8	1
9	23
10	34
11	12
12	1
13	23
14	34
15	12
16	1
17	23
18	34
19	12
20	1
21	23

k
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21

d	d * e	mod phi (n)
2	14	14
3	21	21
4	28	28
5	35	35
6	42	2
7	49	9
8	56	16
9	63	23
10	70	30
11	77	37
12	84	4
13	91	11
14	98	18
15	105	25
16	112	32
17	119	39
18	126	6
19	133	13
20	140	20
21	147	27
22	154	34
23	161	1
24	168	8
25	175	15

k	C^k mod n (letter 2)
1	25
2	20
3	5
4	15
5	45
6	25
7	20
8	5
9	15
10	45
11	25
12	20
13	5
14	15
15	45
16	25
17	20
18	5
19	15
20	45
21	25
22	20
23	5

k	C^k mod n (letter 3)
1	23
2	34
3	12
4	1
5	23
6	34
7	12
8	1
9	23
10	34
11	12
12	1
13	23
14	34
15	12
16	1
17	23
18	34
19	12
20	1
21	23
22	34
23	12

k	C^k mod n (letter 4)
1	36
2	31
3	16
4	26
5	1
6	36
7	31
8	16
9	26
10	1
11	36
12	31
13	16
14	26
15	1
16	36
17	31
18	16
19	26
20	1
21	36
22	31
23	16

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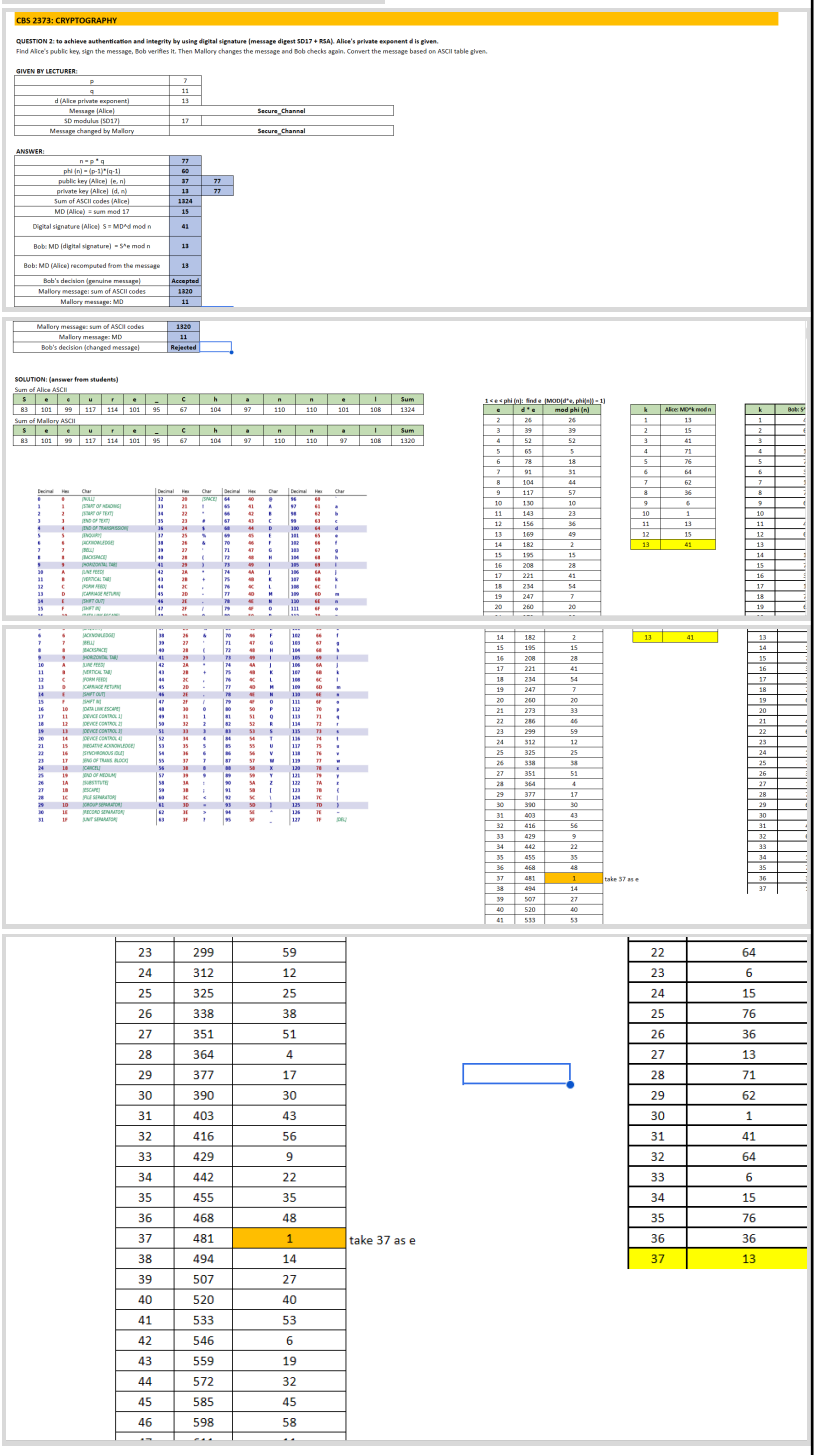
Digital signature: To achieve authentication and integrity by using digital signature when the private key is given, and to detect a changed message (include the calculation)	**QUESTION 2:** Alice's private exponent d is given**. Find Alice's public key,** then sign the message using the SD17 idea (sum of the ASCII codes mod 17) and RSA. Bob verifies the message. Then Mallory changes the message and Bob checks again. Convert each character according to the ASCII table below. Give the answer by using **Excel including the calculation**. **(keys 5 marks + digital signature 7 marks + Bob's verification 8 marks)**	**20**

Decimal	Hex	Char	Decimal	Hex	Char	Decimal	Hex	Char	Decimal	Hex	Char
0	0	[NULL]	32	20	[SPACE]	64	40	@	96	60	`
1	1	[START OF HEADING]	33	21	!	65	41	A	97	61	a
2	2	[START OF TEXT]	34	22	"	66	42	B	98	62	b
3	3	[END OF TEXT]	35	23	#	67	43	C	99	63	c
4	4	[END OF TRANSMISSION]	36	24	$	68	44	D	100	64	d
5	5	[ENQUIRY]	37	25	%	69	45	E	101	65	e
6	6	[ACKNOWLEDGE]	38	26	&	70	46	F	102	66	f
7	7	[BELL]	39	27	'	71	47	G	103	67	g
8	8	[BACKSPACE]	40	28	(	72	48	H	104	68	h
9	9	[HORIZONTAL TAB]	41	29	)	73	49	I	105	69	i
10	A	[LINE FEED]	42	2A	*	74	4A	J	106	6A	j
11	B	[VERTICAL TAB]	43	2B	+	75	4B	K	107	6B	k
12	C	[FORM FEED]	44	2C	,	76	4C	L	108	6C	l
13	D	[CARRIAGE RETURN]	45	2D	-	77	4D	M	109	6D	m
14	E	[SHIFT OUT]	46	2E	.	78	4E	N	110	6E	n
15	F	[SHIFT IN]	47	2F	/	79	4F	O	111	6F	o
16	10	[DATA LINK ESCAPE]	48	30	0	80	50	P	112	70	p
17	11	[DEVICE CONTROL 1]	49	31	1	81	51	Q	113	71	q
18	12	[DEVICE CONTROL 2]	50	32	2	82	52	R	114	72	r
19	13	[DEVICE CONTROL 3]	51	33	3	83	53	S	115	73	s
20	14	[DEVICE CONTROL 4]	52	34	4	84	54	T	116	74	t
21	15	[NEGATIVE ACKNOWLEDGE]	53	35	5	85	55	U	117	75	u
22	16	[SYNCHRONOUS IDLE]	54	36	6	86	56	V	118	76	v
23	17	[ENG OF TRANS. BLOCK]	55	37	7	87	57	W	119	77	w
24	18	[CANCEL]	56	38	8	88	58	X	120	78	x
25	19	[END OF MEDIUM]	57	39	9	89	59	Y	121	79	y
26	1A	[SUBSTITUTE]	58	3A	:	90	5A	Z	122	7A	z
27	1B	[ESCAPE]	59	3B	;	91	5B	[	123	7B	{
28	1C	[FILE SEPARATOR]	60	3C	<	92	5C	\	124	7C	|
29	1D	[GROUP SEPARATOR]	61	3D	=	93	5D	]	125	7D	}
30	1E	[RECORD SEPARATOR]	62	3E	>	94	5E	^	126	7E	~
31	1F	[UNIT SEPARATOR]	63	3F	?	95	5F	_	127	7F	[DEL]

p	7
q	11
d (Alice private exponent)	13
Message (Alice)	Secure_Channel
SD modulus	17
Message changed by Mallory	Secure_Channal
n and phi (n)	7 & 60
Public key (Alice) (e, n)	?
Alice: sum of ASCII codes and MD	?
Digital signature (Alice)	?
Bob: MD (digital signature)	?
Bob's decision (genuine message)	?
Mallory's message: sum of ASCII codes and MD	?
Bob's decision (changed message), with reason	?

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Screen shot the Excel result:



CBS 2373: CRYPTOGRAPHY

QUESTION 2: to achieve authentication and integrity by using digital signature (message digest SD17 + RSA). Alice's private exponent d is given.
Find Alice's public key, sign the message, Bob verifies it. Then Mallory changes the message and Bob checks again. Convert the message based on ASCII table given.

GIVEN BY LECTURER:

p	7	
q	11	
d (Alice private exponent)	13	
Message (Alice)	Secure_Channel	
SD modulus (SD17)	17	
Message changed by Mallory	Secure_Channel	

ANSWER:

n = p * q	77	
phi (n) = (p-1)*(q-1)	60	
public key (Alice) (e, n)	37	77
private key (Alice) (d, n)	13	77
Sum of ASCII codes (Alice)	1324	
MD (Alice) = sum mod 17	15	
Digital signature (Alice) S = MD^d mod n	41	
Bob: MD (digital signature) = S^e mod n	13	
Bob: MD (Alice) recomputed from the message	13	
Bob's decision (genuine message)	Accepted	
Mallory message: sum of ASCII codes	1320	
Mallory message: MD	11	

Mallory message: sum of ASCII codes	1320
Mallory message: MD	11
Bob's decision (changed message)	Rejected

SOLUTION: (answer from students)

Sum of Alice ASCII

S	e	c	u	r	e	_	C	h	a	n	n	e	l	Sum
83	101	99	117	114	101	95	67	104	97	110	110	101	108	1324

Sum of Mallory ASCII

S	e	c	u	r	e	_	C	h	a	n	n	e	l	Sum
83	101	99	117	114	101	95	67	104	97	110	110	97	108	1320

1 < e < phi (n): find e (MOD(d*e, phi(n)) = 1)

e	d * e	mod phi (n)
2	26	26
3	39	39
4	52	52
5	65	5
6	78	18
7	91	31
8	104	44
9	117	57
10	130	10
11	143	23
12	156	36
13	169	49
14	182	2
15	195	15
16	208	28
17	221	41
18	234	54
19	247	7
20	260	20
21	273	33
22	286	46
23	299	59
24	312	12
25	325	25
26	338	38
27	351	51
28	364	4
29	377	17
30	390	30
31	403	43
32	416	56
33	429	9
34	442	22
35	455	35
36	468	48
37	481	1
38	494	14
39	507	27
40	520	40
41	533	53

take 37 as e

k	Alice: MD^k mod n
1	13
2	15
3	41
4	71
5	76
6	64
7	62
8	36
9	6
10	1
11	13
12	15
13	41

k	Bob: M
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
12	
13	
14	
15	
16	
17	
18	
19	

23	299	59	
24	312	12	
25	325	25	
26	338	38	
27	351	51	
28	364	4	
29	377	17	
30	390	30	
31	403	43	
32	416	56	
33	429	9	
34	442	22	
35	455	35	
36	468	48	
37	481	1	take 37 as e
38	494	14	
39	507	27	
40	520	40	
41	533	53	
42	546	6	
43	559	19	
44	572	32	
45	585	45	
46	598	58	

k	Alice: MD^k mod n
22	64
23	6
24	15
25	76
26	36
27	13
28	71
29	62
30	1
31	41
32	64
33	6
34	15
35	76
36	36
37	13

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Digital certificate:

To verify digital certificates from a Certificate Authority (CA) and to detect a forged certificate (include the calculation)

QUESTION 3: **15**

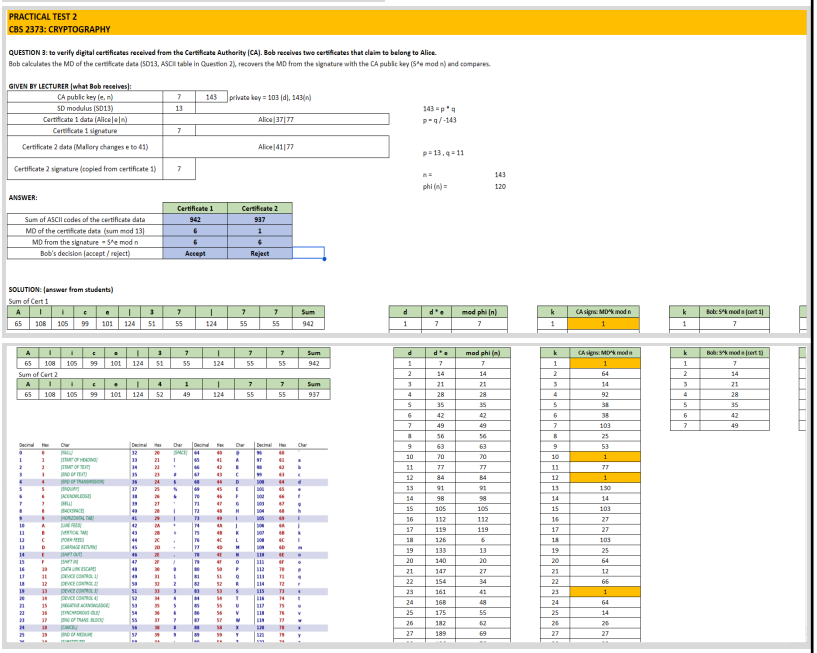
Bob receives two digital certificates that claim to belong to Alice. Both are signed by the Certificate Authority (CA). Use the SD13 idea for the message digest (ASCII table in Question 2). Bob calculates the MD of the certificate data, recovers the MD from the signature with the CA public key (S^e mod n) and compares. Give the answer by using **Excel including the calculation**.

(Certificate 1: 7.5 marks + Certificate 2: 7.5 marks)

CA public key (e, n)	7, 143
SD modulus	13
Certificate 1 data	Alice|37|77
Certificate 1 signature	7
Certificate 2 data	Alice|41|77
Certificate 2 signature	7

	Certificate 1	**Certificate 2**
Sum of ASCII codes	?	?
MD of certificate data	?	?
MD from signature (S^e mod n)	?	?
Bob's decision, with reason	?	?

Screen shot the Excel result:



PRACTICAL TEST 2
CBS 2373: CRYPTOGRAPHY

QUESTION 3: to verify digital certificates received from the Certificate Authority (CA). Bob receives two certificates that claim to belong to Alice.
Bob calculates the MD of the certificate data (SD13, ASCII table in Question 2), recovers the MD from the signature with the CA public key (S^e mod n) and compares.

GIVEN BY LECTURER (what Bob receives):

CA public key (e, n)	7	143	private key = 103 (d), 143(n)
SD modulus (SD13)	13		
Certificate 1 data (Alice|e|n)	Alice|37|77		
Certificate 1 signature	7		
Certificate 2 data (Mallory changes e to 41)	Alice|41|77		
Certificate 2 signature (copied from certificate 1)	7		

143 = p * q
p = q / -143

p = 13 , q = 11

n = 143
phi (n) = 120

ANSWER:

	Certificate 1	Certificate 2
Sum of ASCII codes of the certificate data	942	937
MD of the certificate data (sum mod 13)	6	1
MD from the signature = S^e mod n	6	6
Bob's decision (accept / reject)	Accept	Reject

SOLUTION: (answer from students)

Sum of Cert 1

A	l	i	c	e	|	3	7	|	7	7	Sum
65	108	105	99	101	124	51	55	124	55	55	942

d	d * e	mod phi (n)
1	7	7
2	14	14
3	21	21
4	28	28
5	35	35
6	42	42
7	49	49
8	56	56
9	63	63
10	70	70
11	77	77
12	84	84
13	91	91
14	98	98
15	105	105
16	112	112
17	119	119
18	126	6
19	133	13
20	140	20
21	147	27
22	154	34
23	161	41
24	168	48
25	175	55
26	182	62
27	189	69

k	CA signs: MD^k mod n
1	1
2	64
3	14
4	92
5	38
6	38
7	103
8	25
9	53
10	1
11	77
12	1
13	130
14	14
15	103
16	27
17	27
18	103
19	25
20	64
21	12
22	66
23	-1
24	64
25	14
26	26
27	27

k	Bob: S^k mod n (cert 1)
1	7
2	14
3	21
4	28
5	35
6	42
7	49

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2	71.5	
3	47.666666	
4	35.75	
5	28.6	
6	23.833333	
7	20.428571	
8	17.875	
9	15.888888	
10	14.3	
11	13	take as p (only integer num)
12	11.916666	
13	11	take as q (only integer num)
14	10.214285	
15	9.5333333	
16	8.9375	
17	8.4117647	
18	7.9444444	
19	7.5263157	
20	7.15	

	99	693	93
	100	700	100
	101	707	107
	102	714	114
take as d	103	721	1
	104	728	8
	105	735	15
	106	742	22
	107	749	29
	108	756	36
	109	763	43

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